

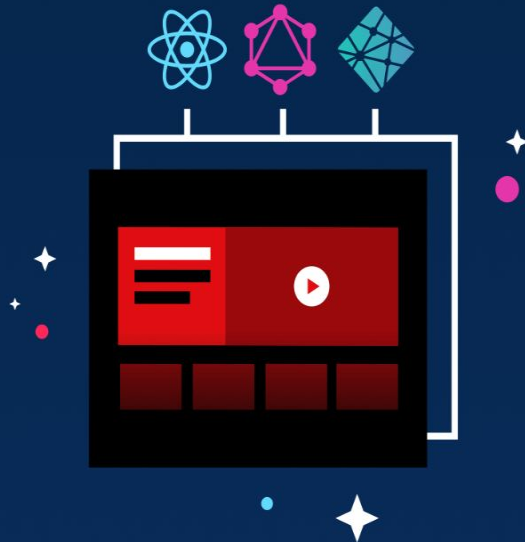
LIVE HANDS-ON WORKSHOP

Build a Netflix clone with GraphQL and React

January 04, Wednesday

8am PT | 4pm GMT | 5pm CET | 9.30pm IST

DataStax



01



Housekeeping

Live and Hands-on

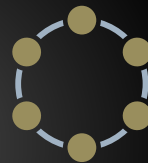
02



Our Goal

Component overview

03



Database Setup

Data model & Astra DB

04



GraphQL

GraphQL

Learn & Hands-On

05



Serverless + Client

Run the application

06



What's next?

Quiz, Homework, ...



Agenda



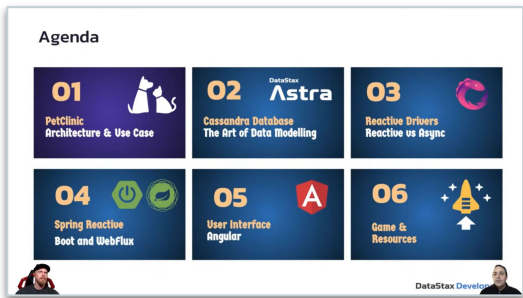


Housekeeping

Live and Hands-on



Livestream: youtube.com/DataStaxDevs



Questions: <https://dtsx.io/discord>



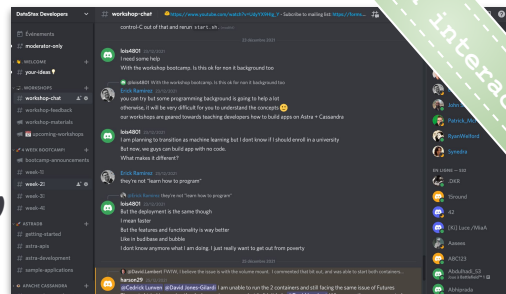
YouTube
(with nightbot)



Discord
(#workshop-chat)

LIVE
CHAT

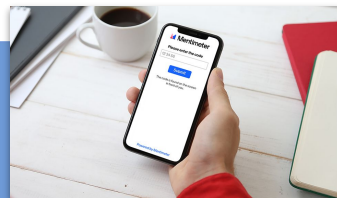
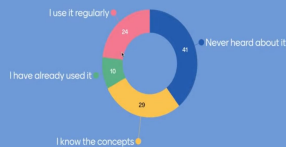
!discord



Games and quizzes: menti.com

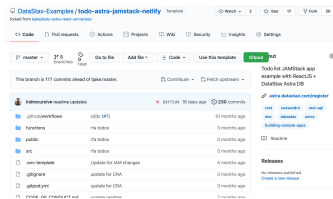


How much experience do you have with the Spring Framework ?

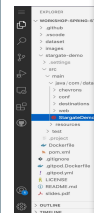


Live Sessions

Code + Slides + Tutorial

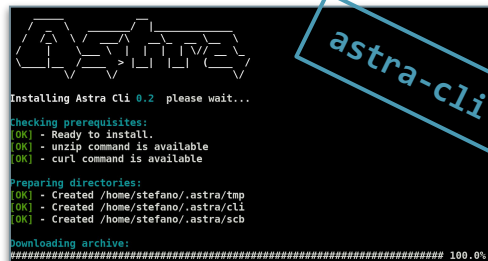


IDE

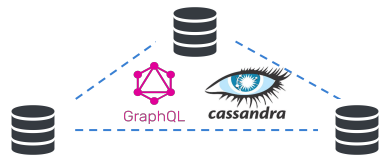


Gitpod

Command-line database management



Database + GraphQL + Playground

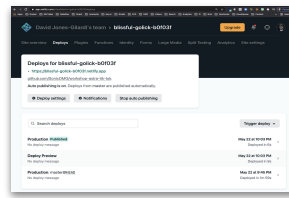


DataStax

Astra DB

STARGATE

Run + Deploy + Host



netlify

Hands-On Housekeeping

Nothing to install!



Get your badge for today

01



Housekeeping

Live and Hands-on

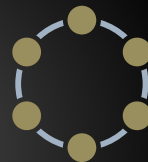
02



Our Goal

Overall architecture

03



Database Setup

Data model & Astra DB

04

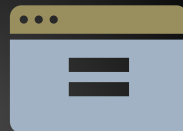


GraphQL

GraphQL

Learn & Hands-On

05



Serverless + Client

Run the application

06



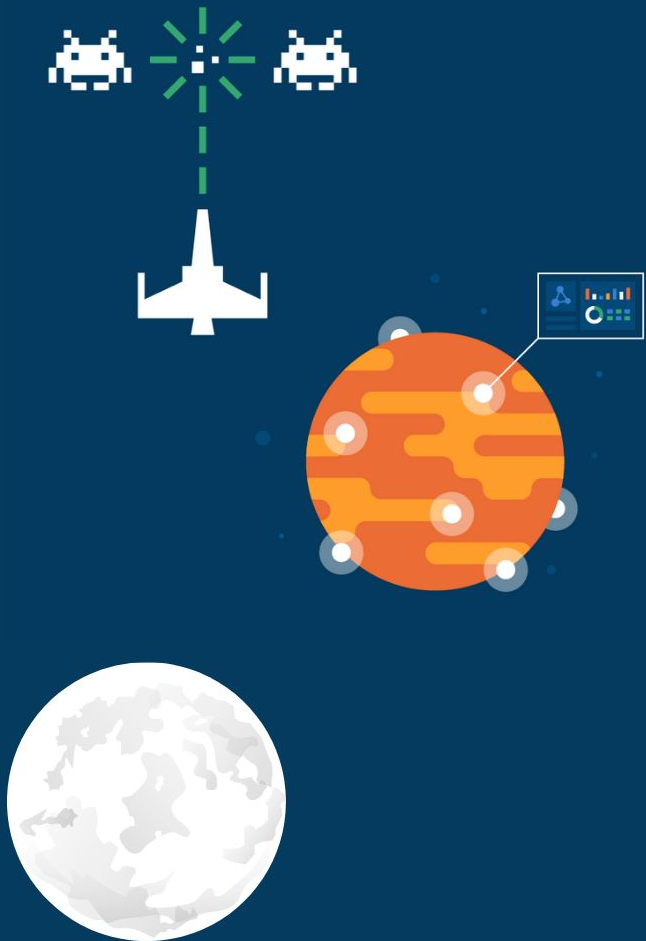
What's next?

Quiz, Homework, ...



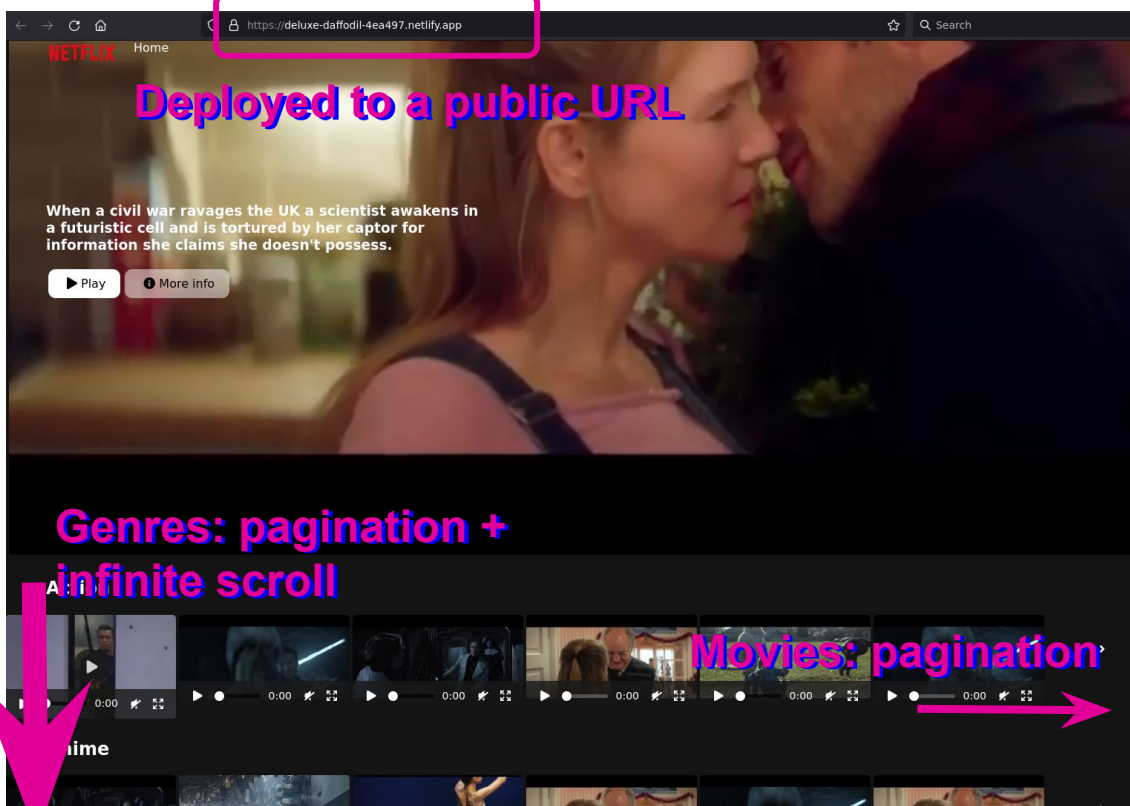
Agenda





Our Goal

Overall Architecture

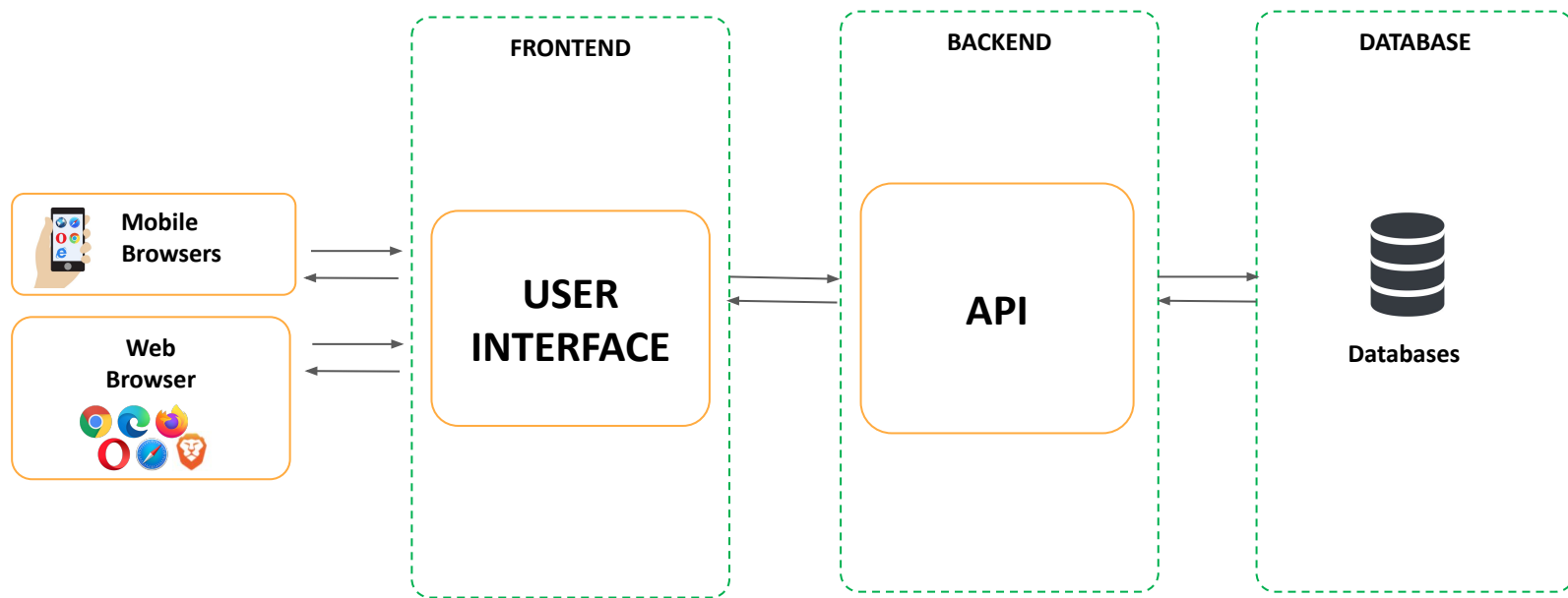


Data
actually
from a DB

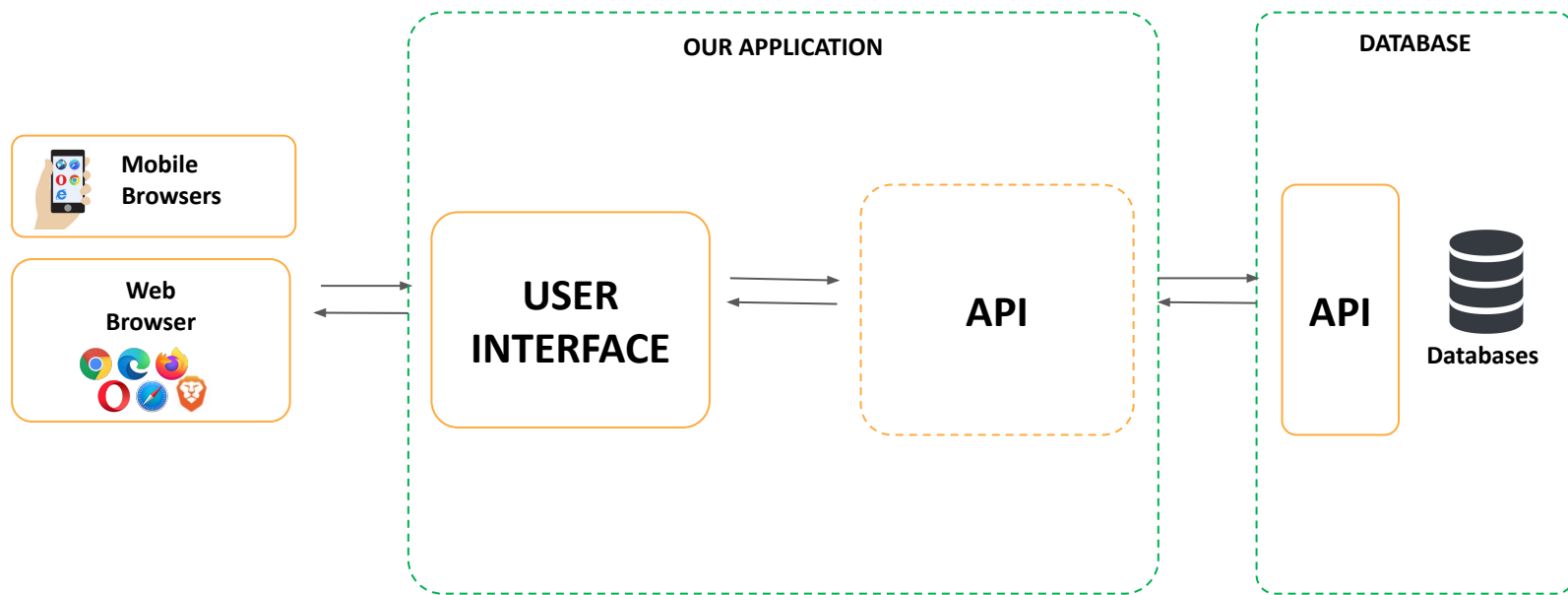


What we want

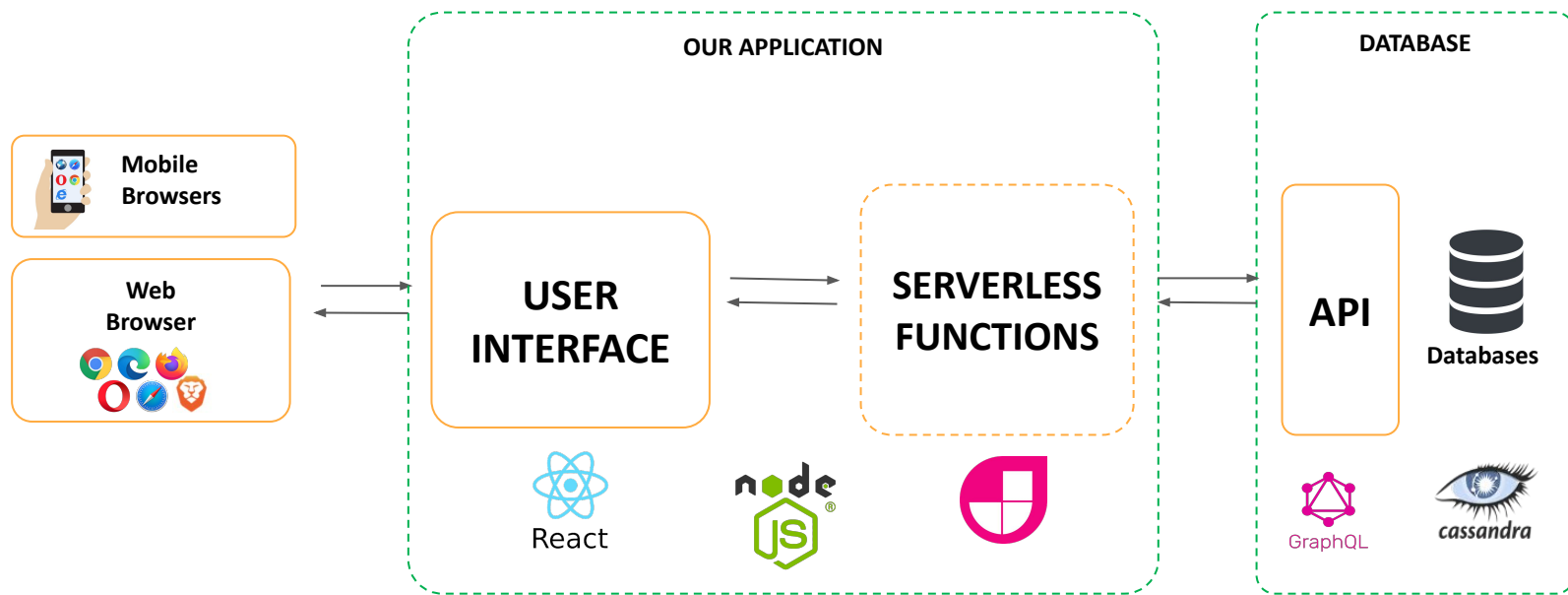




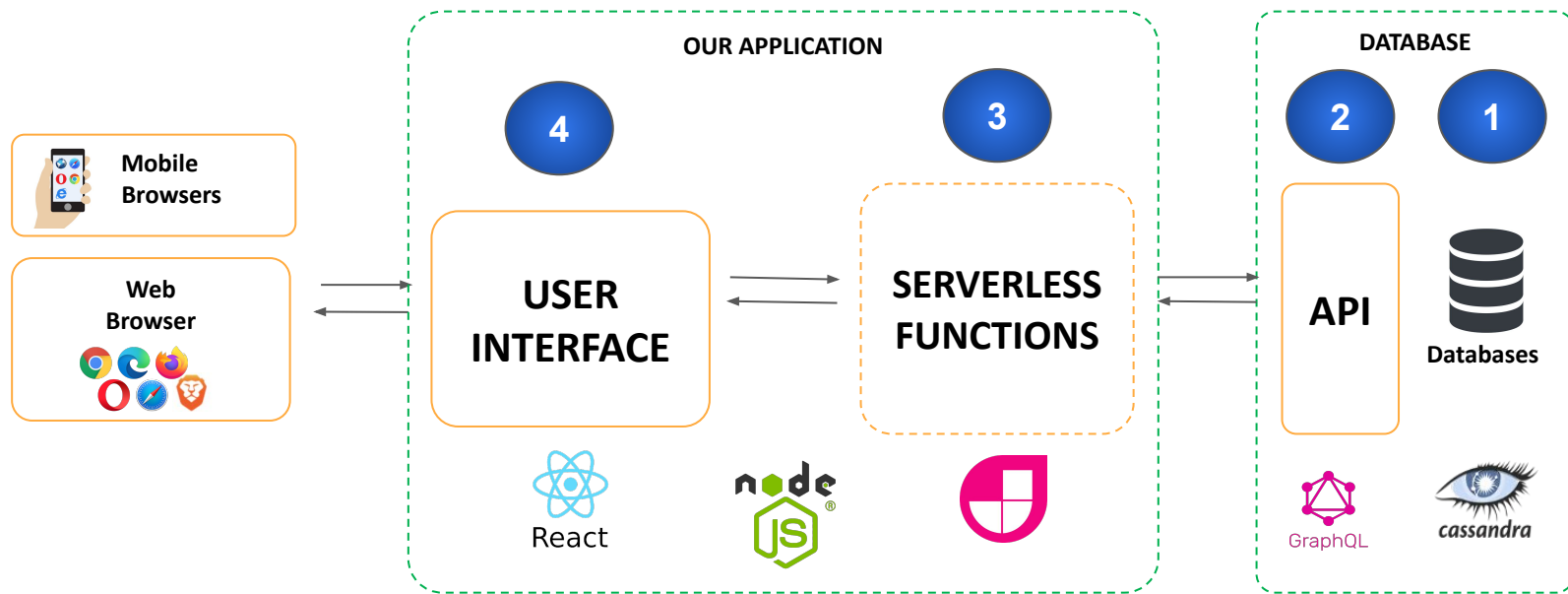
Building a Full-stack application



Building a Full-stack application



Building a Full-stack application



Gitpod



netlify

DataStax
Astra DB

Building a Full-stack application

01



Housekeeping

Live and Hands-on

02



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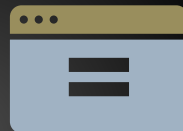


GraphQL

GraphQL

Learn & Hands-On

05



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What's next?

Quiz, Homework, ...



Agenda

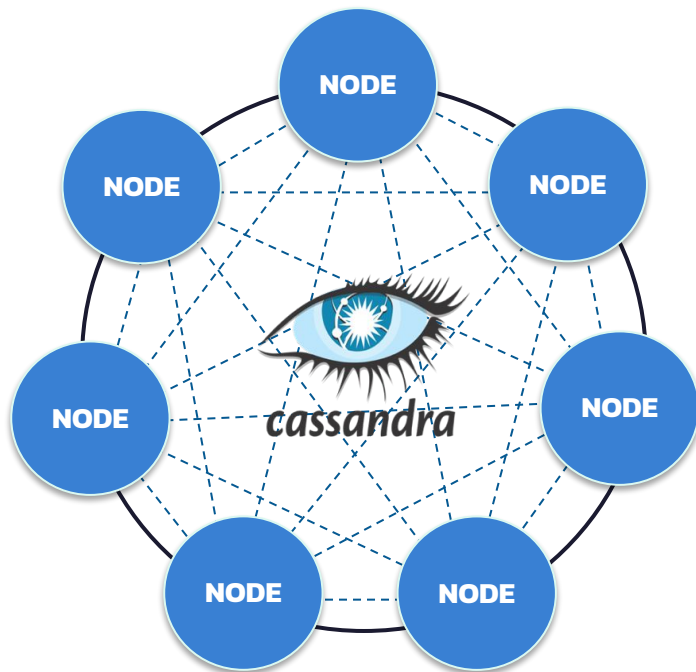


Database setup



Data model & Astra DB





- Big Data Ready
- Read / Write Performance
- Linear Scalability
- Highest Availability
- Self-Healing and Automation
- Geographical Distribution
- Platform Agnostic
- Vendor Independent



Introduction to Apache Cassandra™

CQL API
Compatibility
It is Apache Cassandra !

PULSAR CLIENT
Compatibility
It is Apache Pulsar !

GRAPHQL API
Discoverability, Flexibility
Read only what you need

DOCUMENT API
Flexible Schema
Document-Oriented usages

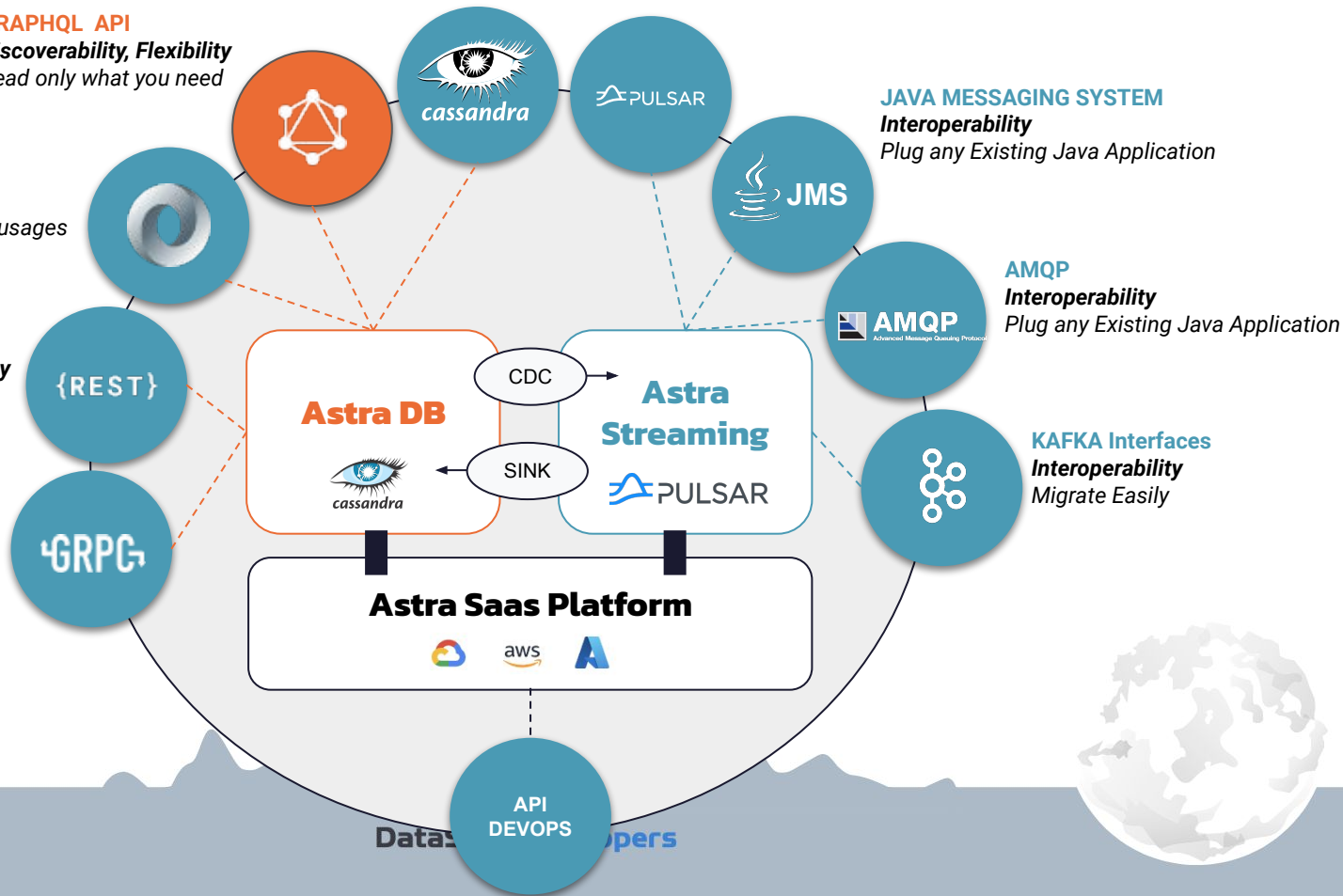
JAVA MESSAGING SYSTEM
Interoperability
Plug any Existing Java Application

AMQP
Interoperability
Plug any Existing Java Application

KAFKA Interfaces
Interoperability
Migrate Easily

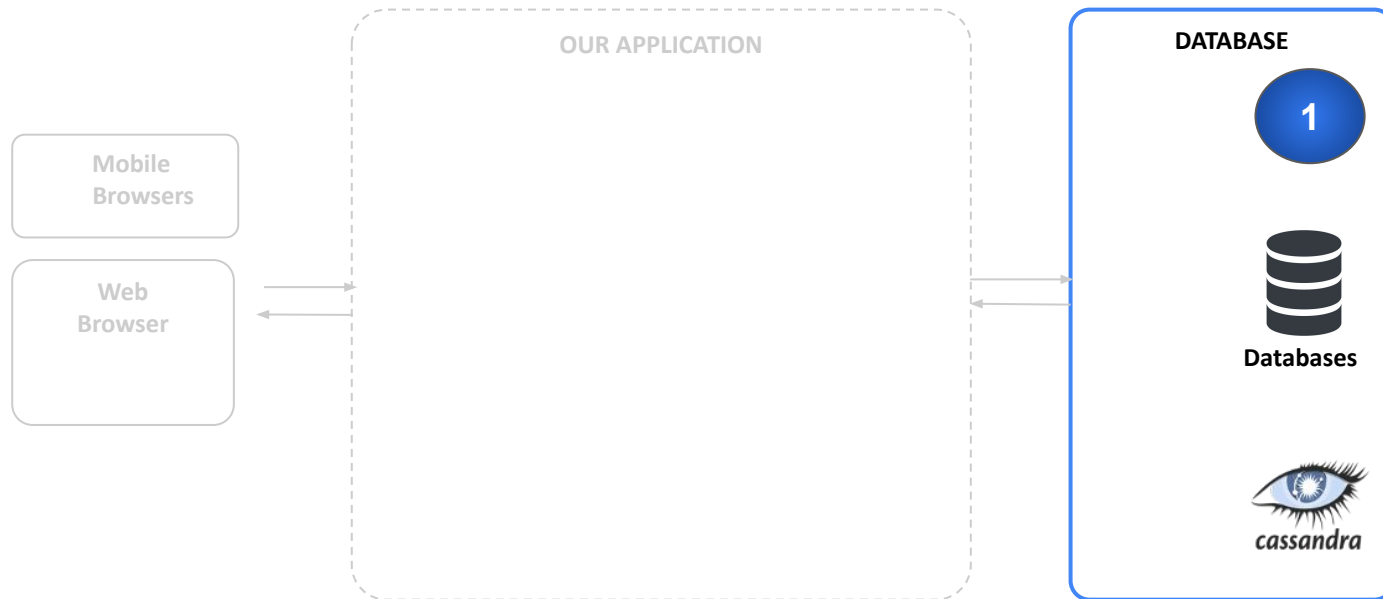
REST API
Stateless, Interoperability
CRUD for Cassandra

GRPC API
Performance, Interoperability
Execute cql in the fastest way



DataStax

Astra DB



Create your Astra DB

Github/datastaxdevs



Create Astra DB

DataStax

Astra DB

OUR APPLICATION

Mobile
Browsers

Web
Browser

DATABASE

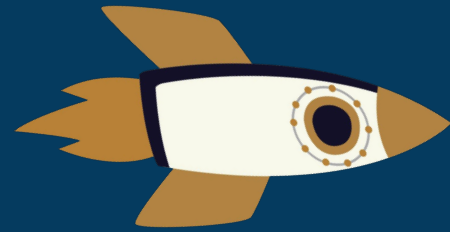
1



Databases



Create your Astra DB



Hands-on (!github)

1. DB SETUP

- ✓ Create (or resume) your Database
- ✓ Create a token

01



Housekeeping

Live and Hands-on

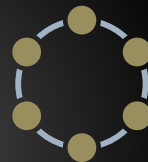
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Database Setup

Data model & Astra DB

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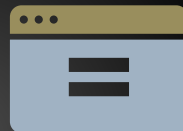


GraphQL

GraphQL

Learn & Hands-On

05



Serverless + Client

Run the application

06



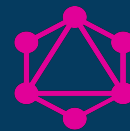
What's next?

Quiz, Homework, ...



Agenda





GraphQL

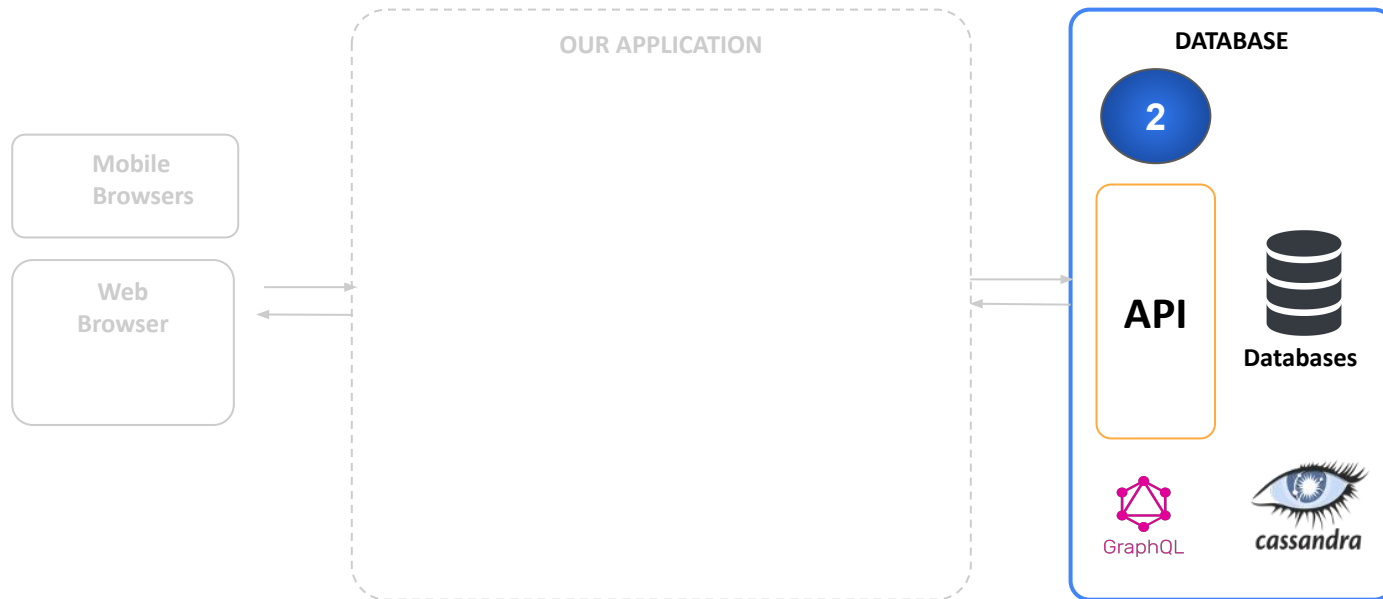
GraphQL

Learn & Hands-on



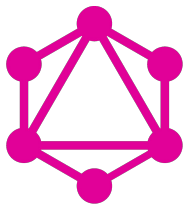
DataStax

Astra DB



GraphQL to access data

- 2012 Created by Facebook
Used internally for mobile apps
- 2015 Facebook give talk at ReactJs Conf
- 2015 Facebook open sourced graphQL
- 2016 Github announced move to GQL



GraphQL

Definition

GraphQL is an application programming interface (API) query language and server-side runtime that prioritises giving customers precisely the data they request.

Meet GraphQL



Describe your data

```
type Project {  
  name: String  
  tagline: String  
  contributors: [User]  
}
```

Ask for what you want

```
{  
  project(name: "GraphQL") {  
    tagline  
  }  
}
```

Get predictable results

```
{  
  "project": {  
    "tagline": "A query language for APIs"  
  }  
}
```



Why GraphQL ?





- ❖ Discoverability, introspection
- ❖ Declarative Data Fetching
- ❖ Schema stitching
- ❖ Customize responses
- ❖ Flexible versioning
- ❖ Match standards (Json | Http)



- ❖ Single endpoint (*versioning, monitoring, security*)
- ❖ Complex implementation (*tooling, still young*)
- ❖ Nice for customers, nasty for DB (*N+1 select*)
- ❖ Rate limiting is required
- ❖ Caching is hard

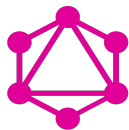
(see Apollo, <https://www.apollographql.com>, used e.g. at Netflix)



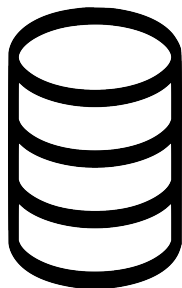
- ❖ BFF : Backend for frontend
- ❖ Aggregation | composition, "federation" (*joins and more*)
- ❖ When bandwidth matters (*mobile phones*)



GraphQL S.W.O.T. analysis



GraphQL



Astra DB

Get the movie genres

get (a pageful of) reference items labeled 'genre'

Get movies for a genre

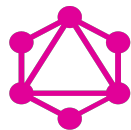
get (a pageful of) movies for a given genre

Data modeling, the Cassandra way:

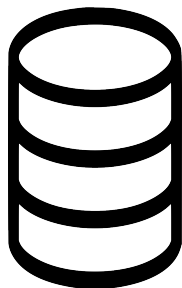
"design the table after the query"

(also: tables are partitioned!)

DB usage patterns



GraphQL



Astra DB

reference_list		
label	text	K
value	text	C ↑

movies_by_genre		
genre	text	K
year	int	C ↓
title	text	C ↑
duration	int	
synopsis	text	
thumbnail	text	

"partition key" (rows stored together)

"primary key" (row uniqueness)

"Chebotko diagrams"

get (a pageful of) reference items labeled 'genre'

get (a pageful of) movies for a given genre

Data modeling

label	value
genre	Action
genre	Anime
genre	Award-Winning
genre	Children & Family
...	

reference_list		
label	text	K
value	text	C↑

genre	year	title	duration
Comedies	2020	AJ and the Queen	142
Comedies	2020	All the Freckles in the World	148
Comedies	2020	Leslie Jones: Time Machine	140
Comedies	2020	Live Twice, Love Once	123
Comedies	2019	"Gabriel ""Fluffy"" Iglesias: One Show Fits All"	134
...			
Reality TV	2019	Fastest Car	117
Reality TV	2019	Flinch	111
Reality TV	2019	Haunted	124
Reality TV	2019	Hyperdrive	144
Reality TV	2019	I'm with the Band: Nasty Cherry	119
Reality TV	2019	Jailbirds	140
...			

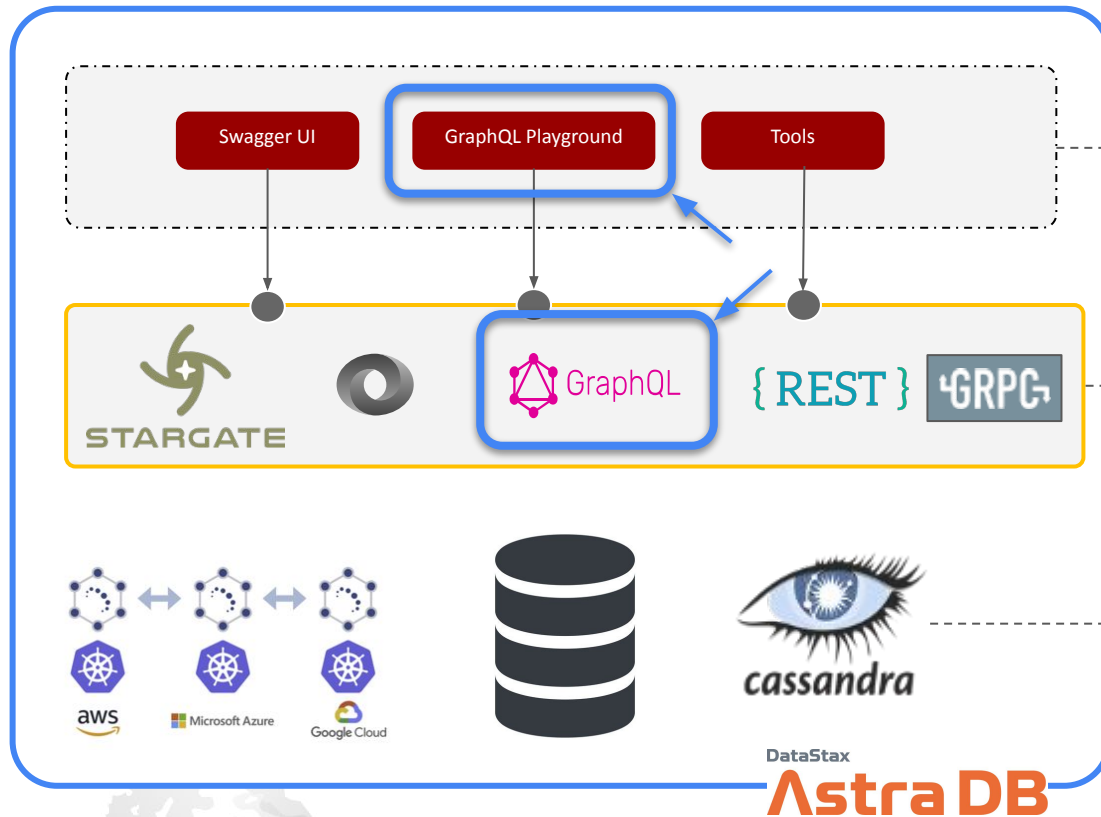
movies_by_genre		
genre	text	K
year	int	C↓
title	text	C↑
duration	int	
synopsis	text	
thumbnail	text	

stored together)

"primary key" (row uniqueness)

"Chebotko diagrams"

Table row examples



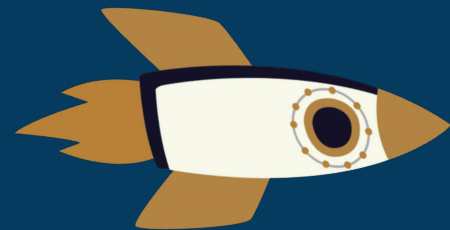
User Interface
Web Utils

OSS Stargate.io
Gateway

Apache Cassandra



Astra DB's GraphQL layer



Hands-on (!github)

2. DATA SETUP



Schema with GraphQL Playground

(*) See next slide!

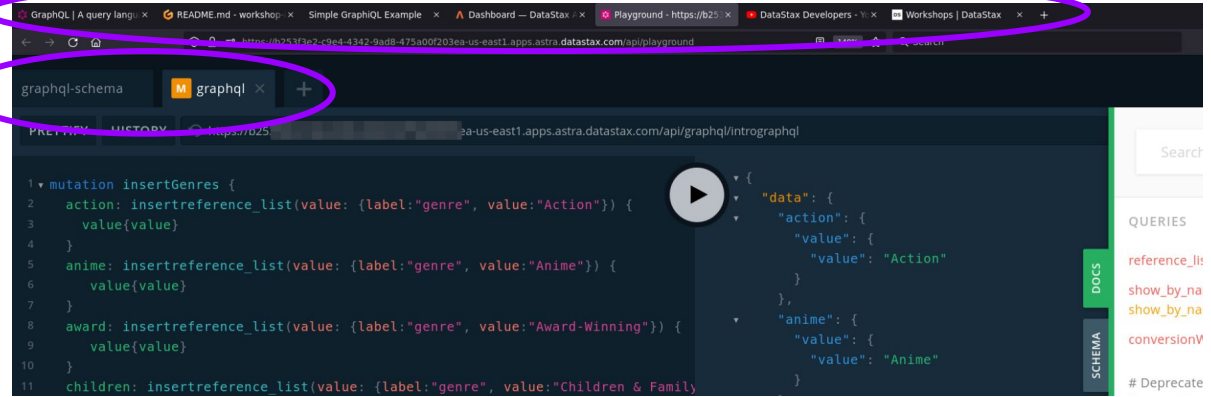


Read/write data with Playground



In your GraphQL Playground ...

These are your browser's tabs



These are the playground tabs ("graphql-schema" & "graphql")



Tabs, tabs, tabs: know the difference

01



Housekeeping

Live and Hands-on

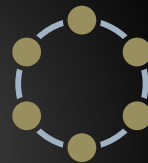
02



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GraphQL

GraphQL

Learn & Hands-On

05



Serverless + Client

Run the application

06



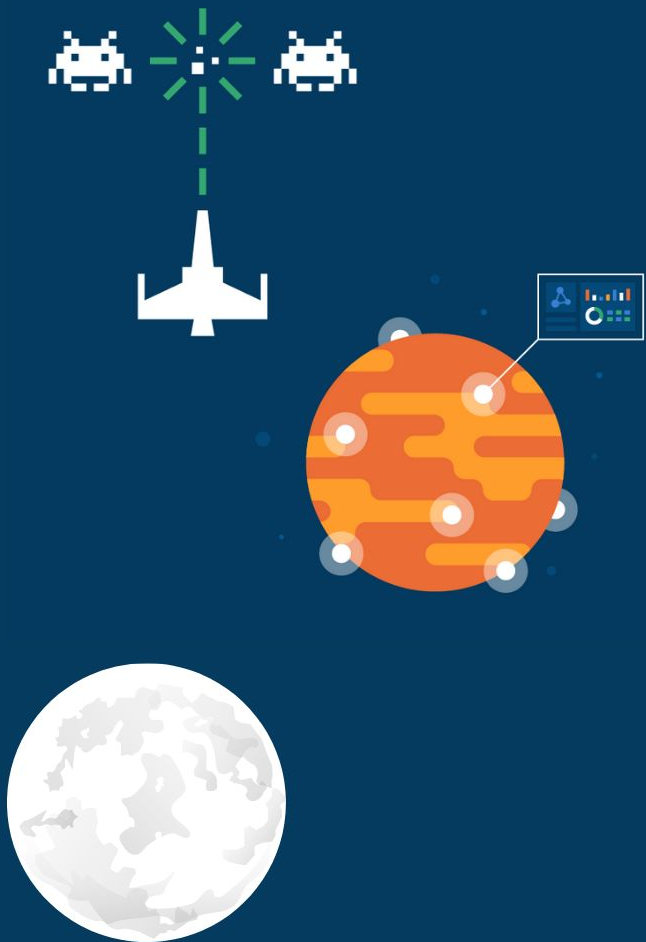
What's next?

Quiz, Homework, ...



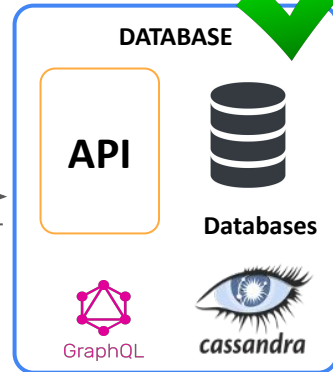
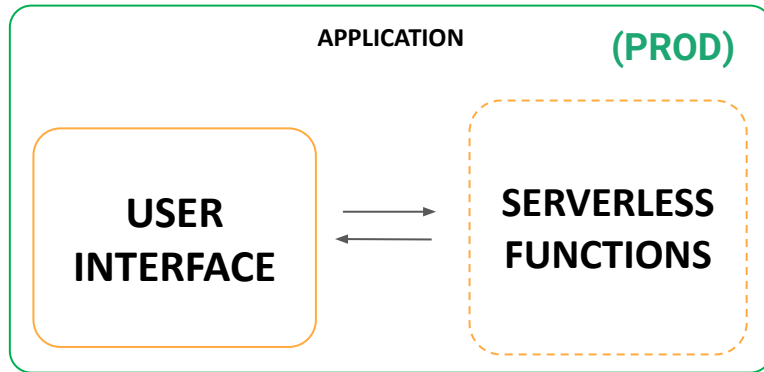
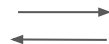
Agenda





Serverless + client

Run the application



Desired prod result



DataStax

Astra DB

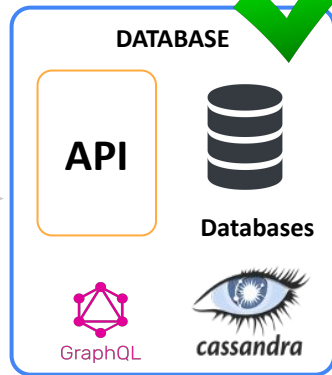


Mobile
Browsers



Web
Browser

Heads up! This slide shows how to JUST get the app running on Gitpod (dev mode), but **it is not the way to Netlify** since you would be running OUR repo in Gitpod instead of a copy in your github. Consult the README for more on this.



Code + Setup + Run Locally



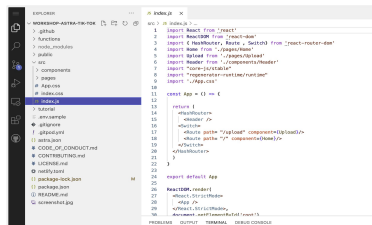
Open in Gitpod

Github/datastaxdevs

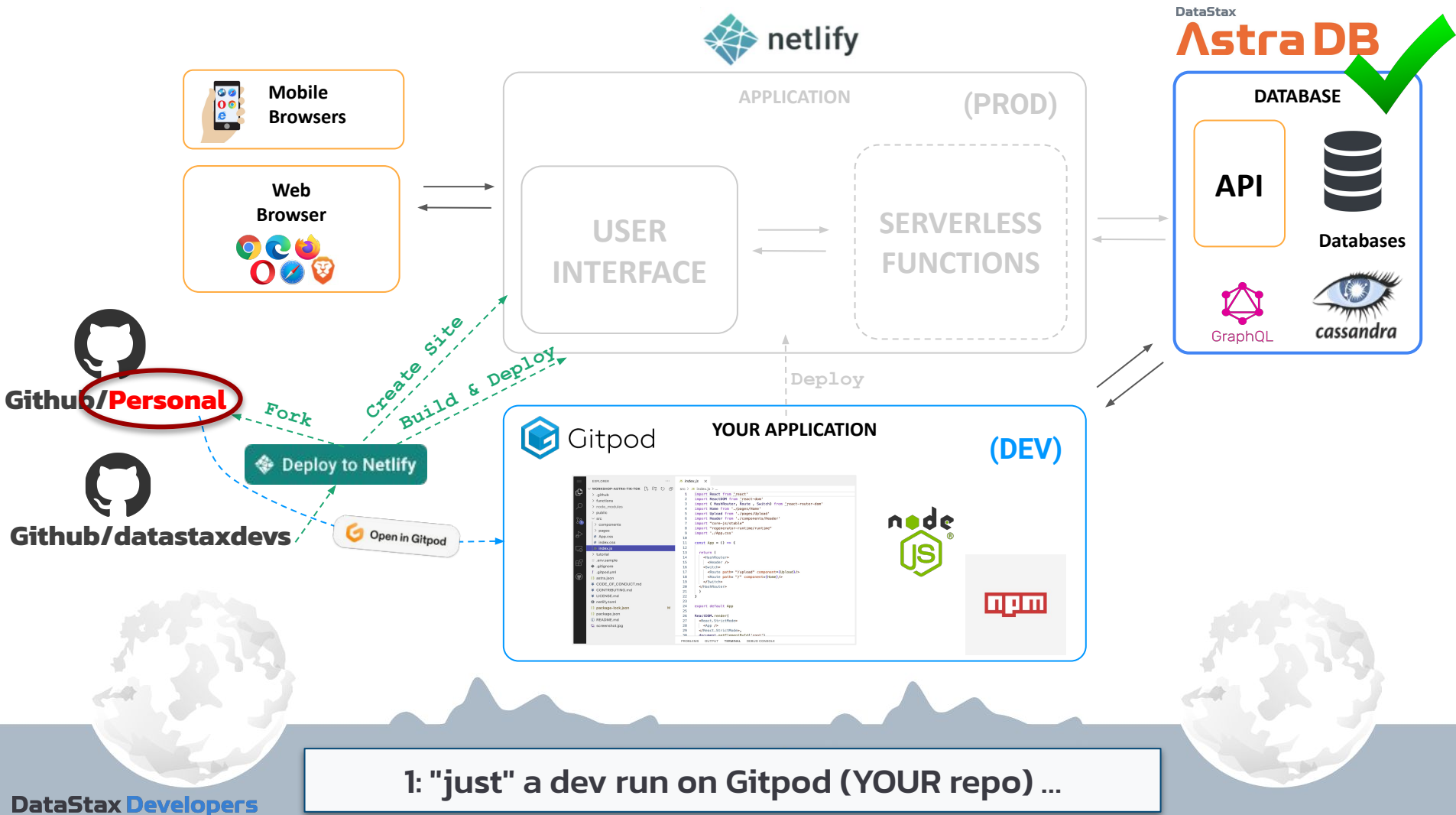


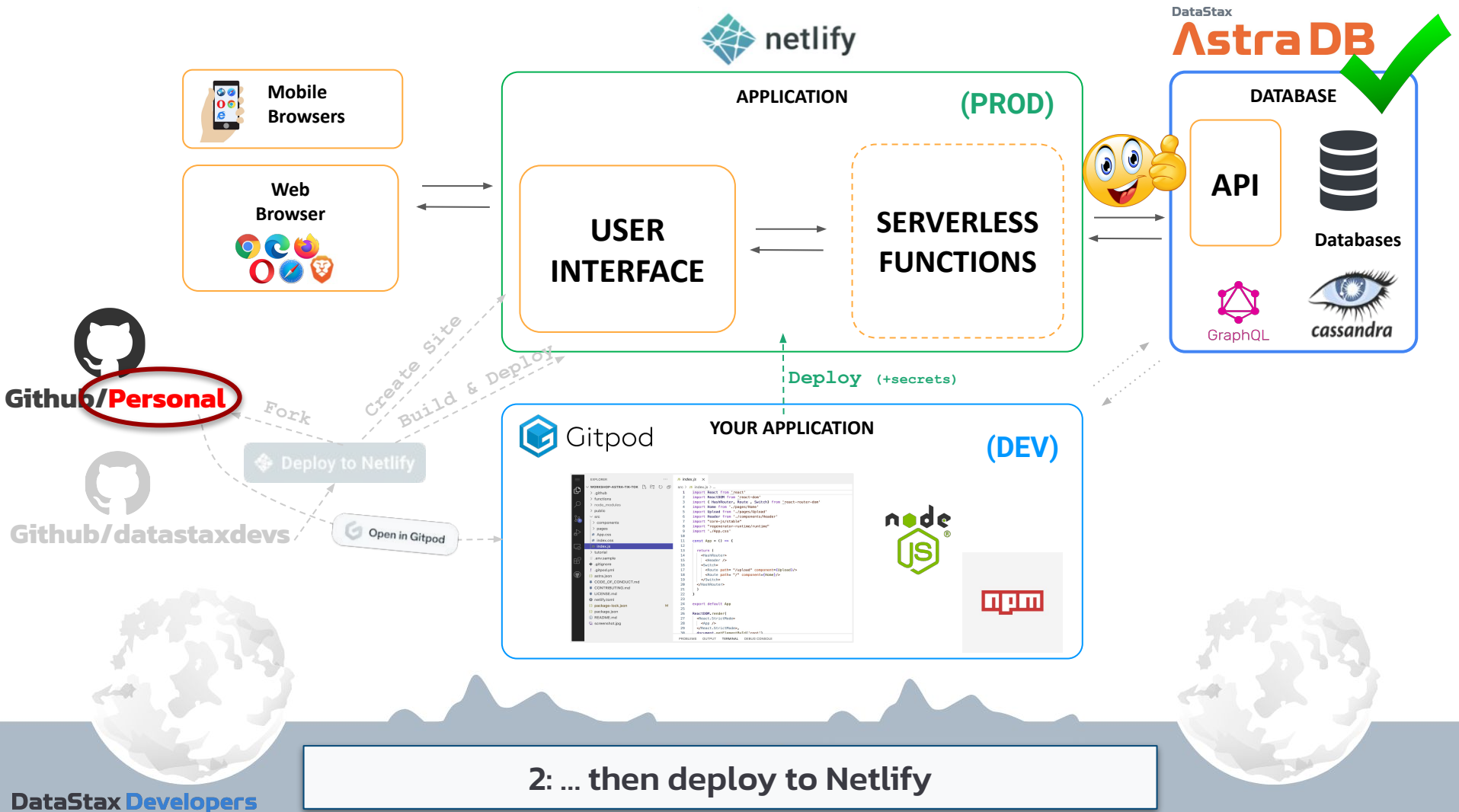
OUR APPLICATION

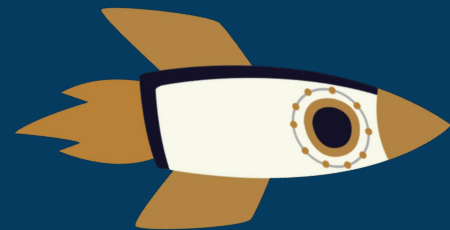
(DEV)



Dev run on Gitpod (no prod deploy in sight)







Hands-on (!github)

3. RUN APP

- ✓ *Copy to YOUR repo and prepare Netlify*
- ✓ Gitpod: bulk data load + dev run
- ✓ Inspect serverless + client code
- ✓ Deploy in production  netlify

01



Housekeeping

Live and Hands-on

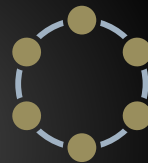
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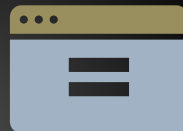


GraphQL

GraphQL

Learn & Hands-On

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What's next?

Quiz, Homework, ...



Agenda



What's next?



Quiz, Homework, ...



- **Homework**

- Complete hands-on by adding YOUR OWN movie to the DB
- Show the NEW movie in the running app (Gitpod is fine)
- Submit screenshot to us ... and wait for your **badge**!

- **Optionally**

- **(Strongly suggested)** deploy to 
- Client code deep dive with Ania's video

- Do reach us for anything: dtsx.io/discord !

All info on Github - !homework



Assignment and Badge



DataStax Developers | **#workshop-chat** | <https://www.youtube.com/watch?v=MuwT5xkFVWI> - Subscribe to mailing list: [http...](http://...)

Left Sidebar (Channels):

- Événements
- # moderator-only
- WELCOME
- start-here
- code-of-conduct
- introductions
- upcoming-events
- useful-resources
- memes
- your-ideas
- the-stage
- WORKSHOPS
- #workshop-chat**
- workshop-feedback
- workshop-materials
- upcoming-workshops
- ASTRADB
- getting-started
- astra-api
- astra-development
- sample-applications
- APACHE CASSANDRA

Channel Content:

RIGGITYREKT Hier à 21:14
I have a 5 node datacenter, 4 nodes are on dse version 5.1.20, one is on dse5.0.15. I am doing some mixed version testing for a class and the one node that is 5.0.15 is coming up as an analytics workload. I dont have /etc/default/dse, instead I am using /etc/init.d/dse-cassandra.
how do i make that node start in cassandra workload, not in analytics?

RIGGITYREKT Hier à 23:39
Okay I found out my issue, when i started DSE 5.0.15 it had endpointsnitch set to DseSimpleSnitch, the rest of my cluster is using PropertyFileSnitch, when i change it to PropertyFileSnitch, it still uses the simple snitch config. looking at the docs i see there is a way to go to GossipingPropertyFileSnitch, but i need the property file one. I can wipe this dbs, do anything with this node to get this done. how do i fix this?
[@here](#)

19 novembre 2021

@RIGGITYREKT Okay I found out my issue, when i started DSE 5.0.15 it had endpointsnitch set to DseSimpleSnitch, the rest...
Erick Ramirez Aujourd'hui à 02:19
mixed versions isn't supported and you're guaranteed to run into weird issues that will cause further problems down the track

@RIGGITYREKT I have a 5 node datacenter, 4 nodes are on dse version 5.1.20, one is on dse5.0.15. I am doing some mixed v...
Cedrick Lunven Aujourd'hui à 09:01
When you start a node you have parameters -k for analytics, -g for graph and -s for search. To remove analytics check and remove -k

Right Sidebar (List of Members):

PRESENTER — 1
David Jones-Gilardi

HELPER — 7
012345
AaronP
BINARY
Chelsea Navo
Jeremy Hanna
John Sanda
Patrick_McFadin

EN LIGNE — 560
-samu-
6304-42J8
Aahlya
Abdurahim
abhi3pathi
Abhiis.s
Abhineet
Abirsh

Bottom Bar: Envoyer un message dans #workshop-chat

!discord
dtsx.io/discord

Datastax Developers Discord (18k+)

Subscribe

Subscribe



Introduction to NoSQL
331 views •
Streamed 1 week ago



Crash Course | Introduction
to Cassandra for Developers
331 views • 1 week ago



Introduction to NoSQL
Databases
3.5K views •
Streamed 1 week ago



Introduction to NoSQL
Databases
10K views •
Streamed 1 week ago



Build your own NETFLIX
clone!
4K views •
Streamed 2 weeks ago



Build your own NETFLIX
clone!
7.4K views •
Streamed 2 weeks ago



Astra Streaming Demo
177 views • 2 weeks ago



Kubernetes Ingress
Management with Traefik...
496 views •
Streamed 2 weeks ago



Build your own TikTok clone!
1.9K views •
Streamed 3 weeks ago



Build your own TikTok
Clone!
4K views •
Streamed 3 weeks ago



How to use the Connect
Driver in Astra DB
113 views • 4 weeks ago



How to use the CQL Console
in Astra DB
39 views • 4 weeks ago



How to create an
Authentication Token in...
37 views • 4 weeks ago



How to use the Data Loader
in Astra DB
62 views • 4 weeks ago



Astra DB Sample App
Gallery
36 views • 4 weeks ago



How to use Secure Connect
in Astra DB
42 views • 4 weeks ago



Cassandra Day India: CL
Room (Workshops)
2.4K views •
Streamed 4 weeks ago



Cassandra Day India: RF
Room (Talks)
1.3K views •
Streamed 1 month ago

Thank You!

